

Mathematics A

Unit 1 • Chapter OL-T44 Classified

Cross-session • chapter-organised candidate

20 classified items • 85 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

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Contents

Unit 1 • Unit 1 • Chapter OL-T44 • 20 items • 85 marks

Chapter	Items	Marks	Starts
01. OL-T44 Circles, Arcs & Sectors	20	85	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Circles, Arcs & Sectors

Unit 1 • 85 marks • 20 item(s)

June 2025 | 4MA1-1H Q11 | 2 marks

Find sector area

June 2024 | 4MA1-1H Q20 | 4 marks

Find a circular segment area or perimeter

November 2023 | 4MA1-1H Q5 | 3 marks

Find a mixed straight-and-curved perimeter

November 2023 | 4MA1-2H Q20 | 4 marks

Find annular, overlapping or multi-circle area

June 2023 | 4MA1-1H Q22 | 6 marks

Solve an algebraic circle-measure constraint

November 2024 | 4MA1-1H Q16 | 4 marks

Find sector area

November 2024 | 4MA1-2H Q25 | 5 marks

Find annular, overlapping or multi-circle area

June 2022 | 4MA1-1H Q19 | 5 marks

Solve an algebraic circle-measure constraint

June 2022 | 4MA1-2H Q20 | 4 marks

Find annular, overlapping or multi-circle area

January 2022 | 4MA1-1H Q16 | 4 marks

Recover radius or central angle from an arc/sector measure

January 2020 | 4MA1-2H Q11 | 5 marks

Find a mixed straight-and-curved perimeter

January 2021 | 4MA1-1H Q8 | 3 marks

Find a mixed straight-and-curved perimeter

January 2021 | 4MA1-2H Q8 | 5 marks

Find annular, overlapping or multi-circle area

January 2021 | 4MA1-2H Q22 | 6 marks

Solve an algebraic circle-measure constraint

June 2021 | 4MA1-1H Q12 | 2 marks

Find arc length

... 5 more item(s) follow

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Working Unit 08

4MA1-1H Q11

UNIT 1
2 marks

- 11 $OABC$ is a sector of a circle, centre O

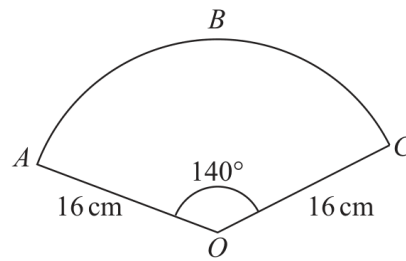


Diagram **NOT**
accurately drawn

$$\text{angle } AOC = 140^\circ$$

$$OA = OC = 16\text{ cm}$$

Calculate the area of the sector.

Give your answer correct to 3 significant figures.

..... cm^2

(Total for Question 11 is 2 marks)

WORKING SPACE

Question 14

4MA1-1H Q20

MODULAR UNIT 1

4 marks

- 20 The diagram shows a sector $OABC$ of a circle centre O

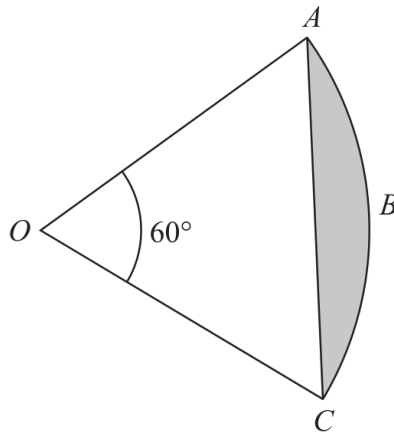


Diagram **NOT**
accurately drawn

Angle $AOC = 60^\circ$

The area of the shaded segment ABC is 38 cm^2

Work out the perimeter of the shaded segment ABC
Give your answer correct to one decimal place.

..... cm

Working space for Question 14

Continue your method

UNIT 1
4 marks

WORKING SPACE

Question 08

4MA1-1H Q5 demand

MODULAR UNIT 1

3 marks

- 5 The diagram shows a shaded shape made from three identical semicircles, *X*, *Y* and *Z*

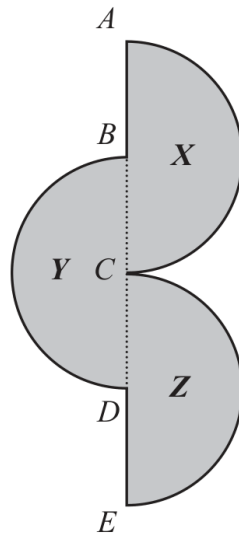


Diagram **NOT**
accurately drawn

ABCDE is a straight line.

AC is the diameter of semicircle *X* and *B* is the centre of semicircle *X*

BD is the diameter of semicircle *Y* and *C* is the centre of semicircle *Y*

CE is the diameter of semicircle *Z* and *D* is the centre of semicircle *Z*

$$AC = BD = CE = 20 \text{ cm}$$

Work out the perimeter of the shaded shape.

Give your answer correct to the nearest whole number.

..... cm

WORKING SPACE

Working space for Question 08

Continue your method

UNIT 1
3 marks

WORKING SPACE

Question 30

4MA1-2H Q20 whole

MODULAR UNIT 1

4 marks

- 20 The diagram shows equilateral triangle ABC with sides of length 10 cm. A circle is drawn inside the triangle.

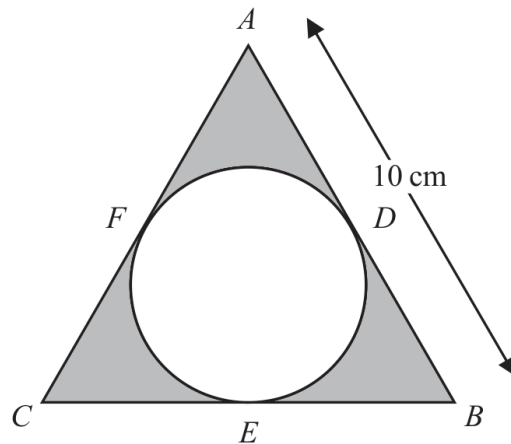


Diagram **NOT**
accurately drawn

D , E and F are points on the circle.

ADB , BEC and CFA are tangents to the circle.

Calculate the total area of the regions shown shaded in the diagram.

Give your answer correct to 3 significant figures.

..... cm^2

WORKING SPACE

Working space for Question 30

Continue your method

UNIT 1
4 marks

WORKING SPACE

Question 11

4MA1-1H Q22

MODULAR UNIT 1

6 marks

22 The diagram shows two circles with centre O and a regular pentagon $ABCDE$

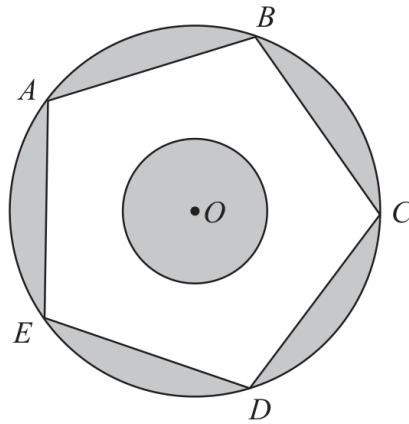


Diagram **NOT**
accurately drawn

A , B , C , D and E are points on the larger circle.
The pentagon has sides of length 8 cm.

The diagram is shaded such that

$$\text{shaded area} = \text{unshaded area}$$

Work out the radius of the smaller circle.
Give your answer correct to 3 significant figures.

..... cm

Working space for Question 11

Continue your method

UNIT 1
6 marks

WORKING SPACE

Question 12

4MA1-1H Q16 M01

MODULAR UNIT 1

4 marks

16 $OAPB$ is a sector of a circle, centre O

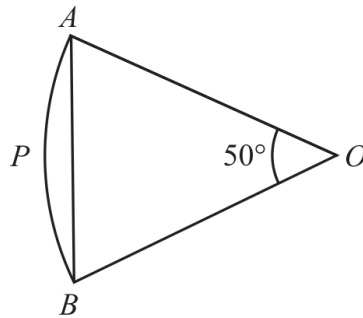


Diagram **NOT**
accurately drawn

Angle $AOB = 50^\circ$

Area of triangle $OAB = 120 \text{ cm}^2$

Work out the area of the sector $OAPB$

Give your answer correct to 3 significant figures.

..... cm^2

WORKING SPACE

Question 28

4MA1-2H Q25 M01

MODULAR UNIT 1

5 marks

25 The diagram shows an equilateral triangle ABC and a circle with centre O

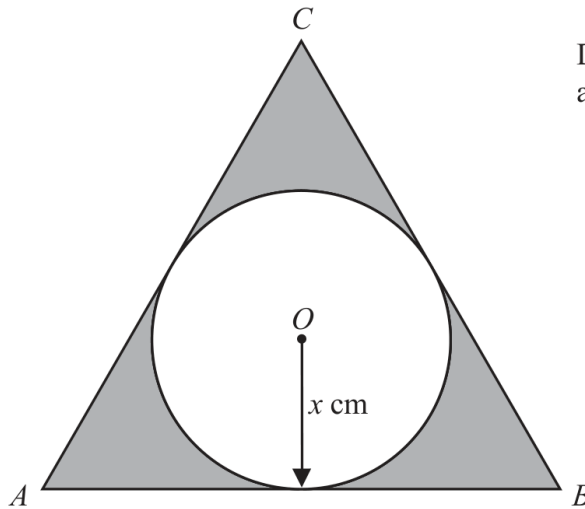


Diagram **NOT**
accurately drawn

AB , BC and CA are tangents to the circle.

The radius of the circle is x cm

The total area, in cm^2 , of the regions shown shaded in the diagram is nx^2

Find the value of n

Give your answer correct to 3 significant figures.

Question 28 (continued)

4MA1-2H Q25 M01

MODULAR UNIT 1

5 marks

 $n = \dots\dots\dots$

WORKING SPACE

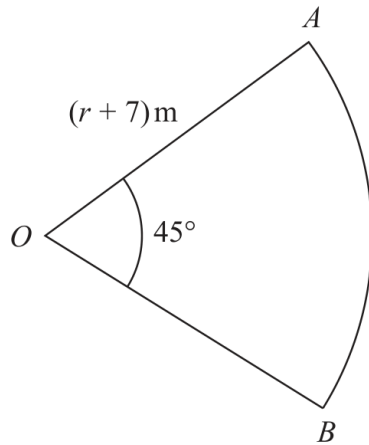
Question 13

4MA1-1H Q19 Q19

MODULAR UNIT 1

5 marks

19

Diagram **NOT**
accurately drawn

OAB is a sector S of a circle with centre O and radius $(r + 7)$ metres.
Angle $AOB = 45^\circ$

A circle C has radius $(r - 2)$ metres.

The area of sector S is twice the area of circle C

Find the value of r

Show your working clearly.

Question 19 continued. $r = \dots\dots\dots$

Working space for Question 13

Continue your method

UNIT 1**5 marks**

WORKING SPACE

Question 27

4MA1-2H Q20 Q20

MODULAR UNIT 1

4 marks

- 20 The diagram shows four identical circles drawn inside a square.

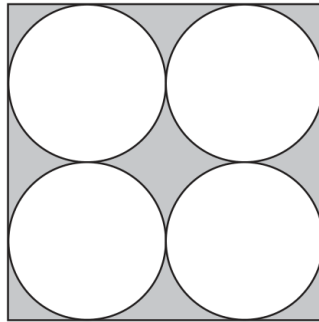


Diagram **NOT**
accurately drawn

Each circle touches two other circles and two sides of the square.

The region inside the square that is outside the circles, shown shaded in the diagram, has a total area of 40 cm^2

Work out the perimeter of the square.

Give your answer correct to 3 significant figures.

..... cm

Working space for Question 27

Continue your method

UNIT 1**4 marks**

WORKING SPACE

Question 14

4MA1-1H Q16

MODULAR UNIT 1

4 marks

16 The diagram shows a circle with centre O

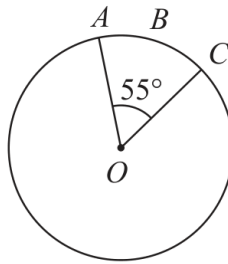


Diagram **NOT**
accurately drawn

A , B and C are points on the circle so that the length of the arc ABC is 5 cm.

Given that angle $AOC = 55^\circ$

work out the area of the circle.

Give your answer correct to one decimal place.

..... cm^2

Working space for Question 14

Continue your method

UNIT 1
4 marks

WORKING SPACE

- 11 The diagram shows a shaded shape $ABCD$ made from a semicircle ABC and a right-angled triangle ACD .

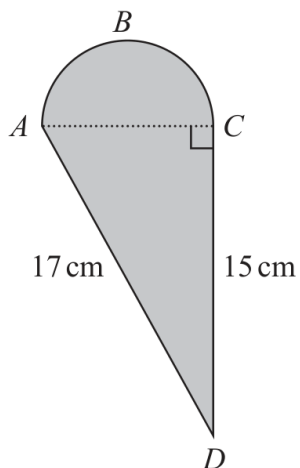


Diagram **NOT**
accurately drawn

AC is the diameter of the semicircle ABC .

Work out the perimeter of the shaded shape.
Give your answer correct to 3 significant figures.

..... cm

YOUR WORKING

- 8 The region, shown shaded in the diagram, is a path.

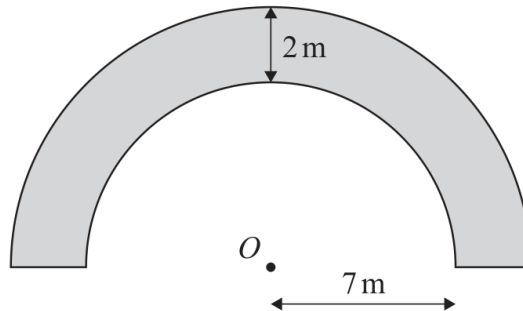


Diagram **NOT**
accurately drawn

The boundary of the path is formed by two semicircles, with the same centre O , and two straight lines.

The inner semicircle has a radius of 7 metres.

The path has a width of 2 metres.

Work out the perimeter of the path.

Give your answer correct to one decimal place.

..... m

YOUR WORKING

8 A , B and C are points on a circle with centre O .

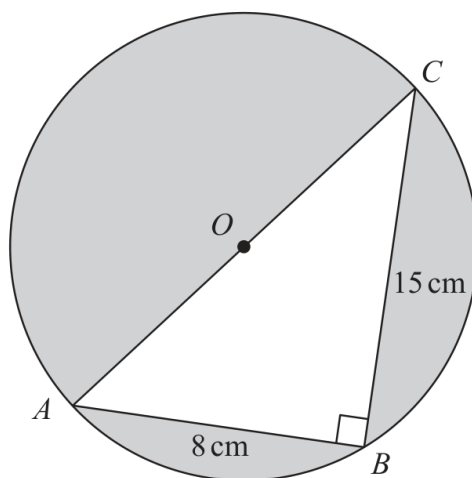


Diagram **NOT**
accurately drawn

AOC is a diameter of the circle.

$AB = 8 \text{ cm}$ $BC = 15 \text{ cm}$

Angle $ABC = 90^\circ$

Work out the total area of the regions shown shaded in the diagram.
Give your answer correct to 3 significant figures.

..... cm^2

YOUR WORKING

22 The diagram shows a sector OBC of a circle with centre O and radius $(6 + x)$ cm.

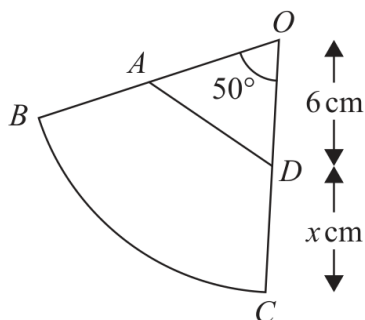


Diagram **NOT**
accurately drawn

A is the point on OB and D is the point on OC such that $OA = OD = 6$ cm

Angle $BOC = 50^\circ$

Given that

the perimeter of sector $OBC = 2 \times$ the perimeter of triangle OAD

find the value of x .

Give your answer correct to 3 significant figures.

$x =$

YOUR WORKING

12 The diagram shows a sector of a circle with radius 7 cm.

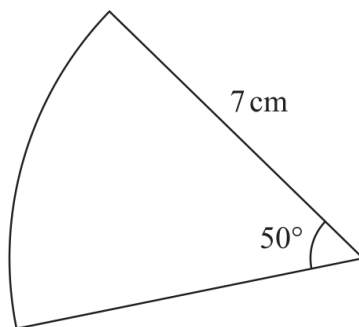


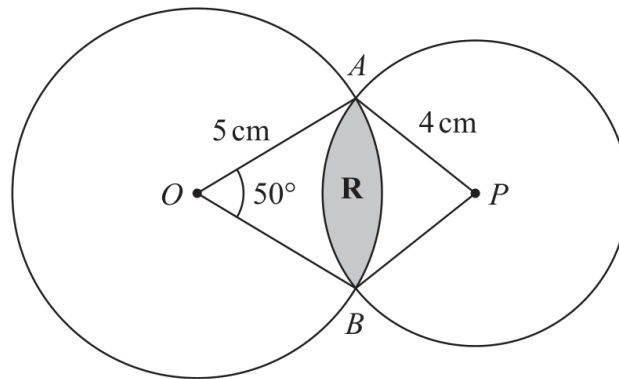
Diagram **NOT**
accurately drawn

Work out the length of the arc of the sector.
Give your answer correct to one decimal place.

..... cm

YOUR WORKING

- 25 The diagram shows two circles such that the region **R**, shown shaded in the diagram, is the region common to both circles.



One of the circles has centre O and radius 5 cm .
 The other circle has centre P and radius 4 cm .
 Angle $AOB = 50^\circ$

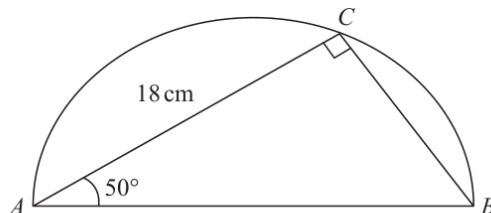
Calculate the area of region **R**.
 Give your answer correct to 3 significant figures.

..... cm^2

YOUR WORKING

- 8 The diagram shows a triangle ABC inside a semicircle.

Diagram **NOT**
accurately drawn



A , B and C are points on the semicircle.

AB is the diameter of the semicircle.

Angle $ACB = 90^\circ$

Angle $BAC = 50^\circ$

$AC = 18$ cm

Work out the perimeter of the semicircle.

Give your answer correct to 2 significant figures.



YOUR WORKING

..... cm

(Total for Question 8 is 5 marks)

YOUR WORKING

21 The diagram shows the cross section of a circular water pipe.

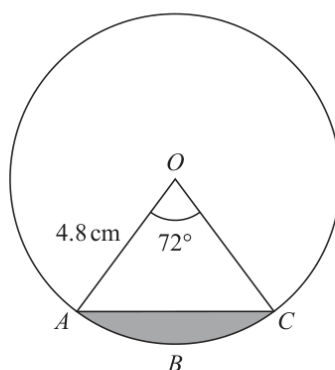


Diagram **NOT**
accurately drawn

OAC is a sector of the circle, centre O

The shaded region in the diagram represents the water flowing in the pipe.

The water flows at 14 cm/s in the pipe.

Work out the volume of water that has flowed through the pipe in 3 minutes.
Give your answer in cm^3 correct to 3 significant figures.



YOUR WORKING

..... cm³

(Total for Question 21 is 5 marks)

YOUR WORKING

15 AOB is a sector of a circle with centre O and radius 12 cm

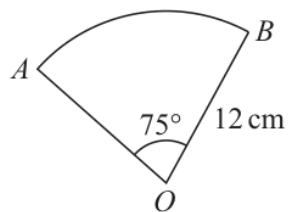


Diagram **NOT**
accurately drawn

Work out the area of the sector.
Give your answer correct to 3 significant figures.

.....cm²

(Total for Question 15 is 2 marks)

YOUR WORKING

- 6 A large circle with centre O contains 3 identical small circles. Each circle touches two other circles.

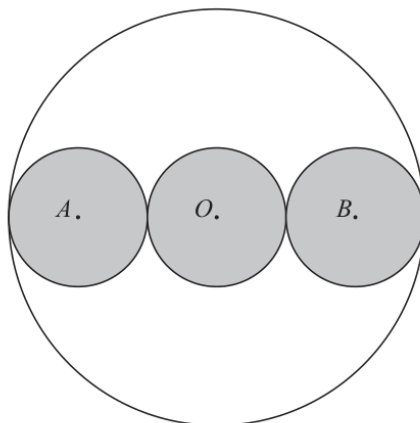


Diagram **NOT**
accurately drawn

A , O and B are the centres of the small circles.
 AOB is a straight line.

The circumference of the large circle is 160 cm

Work out the area of one small circle.
Give your answer correct to 3 significant figures.

..... cm^2

(Total for Question 6 is 5 marks)

YOUR WORKING
